



Department of Computer Science

National University of Computer and Emerging Sciences
Lahore Campus

Data Mining

Section A

Quiz 1b

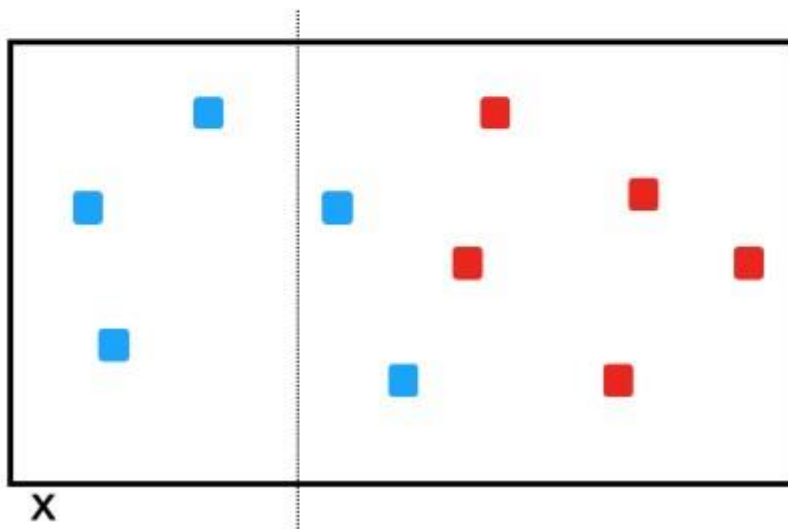
Solution

Question 1: 3 marks

We have used the Decision Tree to classify the dataset below to discriminate blue and red data point. The Decision boundary is shown with a dotted line.

Compute the information gain of this decision boundary.

What is closest information gain of x?



0.39

Question 2: 5 marks

A	B	C	P(A,B,C)
F	F	F	0.008
F	F	T	0.072
F	T	F	0.032
T	F	F	0.012
T	T	F	0.048
F	T	T	0.228
T	F	T	0.108
T	T	T	0.432

- 1) What is $P(A=T | C=T)$
= $P(A = T, C = T) / P(C = T)$ =
= $0.108 + 0.432 / 0.072 + 0.228 + 0.108 + 0.432$
0.64
- 2) What is $P(A=T | B=T)$
= $P(A = T, B = T) / P(B = T)$

1. What is $P(A=T|C=T) = .6$
2. What is $P(A=T|B=T) = .648$

Q 3: 2 marks

True/False

- Variables A and B are independent if knowing A does not change the probability of B. **True**
- Algorithms ID3, which uses information gain as a splitting criterion may have selection bias towards features with a lesser number of distinct values. **False**